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Do extrafloral nectaries present a defensive role against herbivores in two species of the family Bignoniaceae in a Neotropical savanna ?

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Abstract Despite the general belief that the interaction between extrafloral nectaries (EFNs) and ants is mutualistic, the defensive function of EFNs has been poorly documented in South American savannas. In this article, we evaluate the potential impact of EFNs (benefits and costs) on two species of plants from the dry areas of Central Brazil, *Anemopaegma album* and *Anemopaegma scabriusculum* (Bignoniaceae). In particular, we characterize the composition of substances secreted by the EFNs, test whether EFNs attract ants, and whether ants actually present a defensive role, leading to reduced herbivory and increased plant fitness. Histochemical analyses indicated that EFNs from both species of *Anemopaegma* secrete an exudate that is composed of sugars, and potentially lipids and proteins. Furthermore, EFNs from both species were shown to present a significant role in ant attraction. However, contrary to common expectations, ants were not found to protect plants against herbivore attack. No effect was found between ant visitation and flower or fruit production in *A. album*, while the presence of ants led to a significant decrease in flower production in *A. scabriusculum*. These results suggest that EFNs might present a similar cost and benefit in *A. album*, and a higher cost than benefit in *A. scabriusculum*. Since the ancestor of *Anemopaegma* occupied humid forests and already presented EFNs that were maintained in subsequent lineages that occupied savannas, we suggest that phylogenetic inertia might explain the presence of EFNs in the species of *Anemopaegma* in which EFNs lack a defensive function.

Keywords Ant-plant interaction · Context-dependency · Brazil · Caatinga · Cerrado · Natural selection · Nectar glands

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Introduction

Ants and flowering plants interact in a variety of ways (Rico-Gray and Oliveira 2007), and are generally thought to benefit from these interactions (Bronstein 1994). For instance, ants are thought to benefit from the food resources and housing provided by the plants (Beattie 1985), while plants are thought to benefit

from ants for pollination (Hickman 1974), seed dispersal (Christianini and Oliveira 2010), and protection against herbivores (Janzen 1966; quantitative review in Rosumek et al. 2009). However, negative impacts of ants to plants have also been recorded. In particular, the negative effect of leaf-cutter ants to plants (Costa et al. 2008), and the negative effect of granivory (Crist and Macmahon 1992) have been well-documented.

In general, mutual beneficial interactions are mediated by specific morphological structures (e.g., extrafloral nectaries—EFNs) that are concentrated in more vulnerable plant parts (e.g., sprouts, expanding leaves, apical buds, and lateral buds; Heil and McKey 2003). Despite the general belief that EFNs have an important role in plant defense attracting ant bodyguards, it has not been easy to document an increase in plant fitness due to the presence of EFNs (Rico-Gray and Oliveira 2007). It is expected that mutualistic interactions mediated by the EFNs would show "conditionality" or "context-dependency" (Jones and Callaway 2007). Under this assumption, costs and benefits, as well as their respective outcomes, are thought to be affected by current ecological conditions (Bronstein 1994), which would, in turn, explain the potential variations in ant-plant interactions in a predictable way.

In the specific case of ant–plant protection mutualisms, context-dependency does not seem to be a general rule (Chamberlain and Holland 2009), but is thought to be more frequent within species whenever there is variation in local biotic and abiotic conditions (e.g., Tempel 1983; Kersch and Fonseca 2005). In addition, hypotheses that consider the phylogenetic history and geographic context of interactions might also explain the variable outcomes of ant–plant interactions within and between species. For instance, phylogenetic inertia was thought to explain the occurrence of species with EFNs in a Hawaian Island that lacks ants (Keeler 1985). On the other hand, the geographic mosaic theory of coevolution (GMT; Thompson 2005) incorporated a geographical component in biological interaction studies by introducing the idea that different populations from a single species might be under different biotic and abiotic pressures (complex mosaic among populations). Both ideas are directly related to the context-dependency of animal–plant interactions.

The Brazilian savanna covers $\sim$2 million km^2 of Central Brazil (Eiten 1972; Ratter et al. 1997), and includes a great diversity of plants with EFNs ($\sim$15–25% of the woody species according to Oliveira and Leitão-Filho 1987; Oliveira and Brandão 1991). In this biome, EFNs present a variety of morphologies (Machado et al. 2008), and are most abundant in representatives of Leguminosae-Mimosoideae, Bignoniaceae, and Vochysiaceae (Oliveira and Freitas 2004). The defensive hypothesis of EFNs was tested in eight savanna species (Oliveira and Freitas 2004; Rico-Gray and Oliveira 2007; Korndorfer and Del-Claro 2006; Nascimento and Del-Claro 2010; Byk and Del-Claro 2010), however, only half of the studied taxa (*Caryocar brasiliense*, *Chamaecrista debilis*, *Ouratea spectabilis*, and *Qualea multiflora*) presented a defensive role of nectaries, including a negative effect of the EFNs (and associated ants) on herbivory and a positive effect on fitness components (e.g., number of flowers, fruit set, or seeds per fruits).

In the specific case of the plant family Bignoniaceae, the function of EFNs has only been studied in three North American representatives of the Bignoniaceae: *Campsis radicans* (L.) Seem., *Catalpa speciosa* (Warder) Engelm., and *Catalpa bignonioides* Walter (Elias and Gelband 1975; Stephenson 1982; Ness 2003, respectively), leaving a large gap in our current understanding on the role of these structures in the Neotropics. In this study, we used two species of *Anemopaegma* (Bignoniaceae) to test whether EFNs present a defensive function against herbivores. Specifically, we study: (1) The features of the nectar secreted by these two species of *Anemopaegma* and its relationship to substances known to represent rewards to ants; (2) Whether EFNs of *A. album* and *A. scabriusculum* present a role in ant attraction; and (3) Whether ants defend individuals of *Anemopaegma* against herbivore attack and/or whether ants have any impact on flower/fruit production in those species.

Materials and methods

Study area and plants

This study was conducted at the State Park of Grão Mogol and neighboring areas located 550 km north of Belo Horizonte (Minas Gerais, Brazil). The region presents a seasonal climate with temperatures between 20 and 30°C in the rainy season and between 13 and 27°C in the dry season (http://www.inmet.gov.br/).

The mean annual rainfall is around 1,100 mm, with a dry season between May and September. The soil is composed of remnants of ancient crystalline surfaces (Proterozoic) and is predominantly shallow, acidic and poor in nutrients and organic matter (Pirani et al. 2003).

The genus *Anemopaegma* Mart. ex Meisn (Bignonieae, Bignoniaceae) is composed by forty-one species distributed through forests ($N = 30$) and savannas ($N = 11$) from Mexico to Argentina (Lohmann 2003, 2011); and highly supported (100%) as monophyletic (Lohmann 2006). Two species of *Anemopaegma* with great levels of variation in the abundance of foliar EFNs were selected for this study: *A. album* and *A. scabriusculum*.

Anemopaegma album Mart. ex DC. is a shrub distributed throughout Brazilian savannas ('cerrados', 'caatingas', and transitional areas called 'carrascos') of Minas Gerais and Bahia (Brazil), with glabrous to puberulous leaves, and triangular prophylls of the axillary buds. The leaves are trifoliate with EFNs grouped at the axils of veins in the abaxial side of leaflets, and sparsely distributed over the adaxial side of the blade. *Anemopaegma scabriusculum* Mart. ex DC. is a shrub and also occurs throughout the savannas of eastern Brazil, from Minas Gerais to Piauí, reaching the state of Goiás, with pubescent leaves and orbicular prophylls of the axillary buds. The leaves are also trifoliate, but without EFNs grouped. In this species the EFNs are scattered over the adaxial side of the blade, rarely on the abaxial side. To characterize the morphology and exudates of the EFNs in both *Anemopaegma* species we used optical and electronic microscopy, histochemical analyses, urine test strips and a portable refractometer (see Appendix 1S in electronic supplementary material for these detailed methods).

Characterization of ants and herbivore assemblages

Ants and herbivores found in *A. album* and *A. scabriusculum* were collected during field-work and subsequently identified in the laboratory by insect specialist. The behavior of ants found on individuals of *A. album* and *A. scabriusculum* was characterized in the field by describing the degree of use of EFNs, and a description of interactions with herbivores (antagonistic or neutral relationship). Furthermore, the behavior of herbivores was characterized by describing the feeding habit, including in some cases a description of the larval stage in which the herbivores also fed on the plant.

Ant atraction by the EFNs

Observational data were collected on natural populations of *A. album* and *A. scabriusculum* from August 2008 to June 2009. Experimental populations belonging to both species were located in areas of savanna with low human impact, and separated by $\sim$30 km. All plants sampled were taller than 50 cm and included only adult individuals (potentially reproductive according to our observations). Two experiments were conducted to test whether individuals of *A. album* and *A. scabriusculum* attracted more ants than neighboring species, and whether ant abundance found on individuals of *A. album* and *A. scabriusculum* was correlated to the amount of EFNs. These tests were carried out separately on each species.

For the first field experiment, 14 pairs of individuals that included one individual of *A. album* and its nearest heterospecific neighbor, were tagged. Selected heterospecific neighbors were at non-reproductive stage and presented a similar number of leaves than its respective *Anemopaegma* couple. Similarly, 18 pairs of individuals that included one individual of *A. scabriusculum* and its nearest heterospecific neighbor were also tagged. For all tagged individuals of each pair, we recorded the number of ants (abundance) on each experimental plant. We did not exclude neighboring plants with EFNs or honeydew-producing homopterans as we wanted to quantify ant visitation in *A. album* and *A. scabriusculum* under natural conditions (i.e., which also includes situations in which the nearby plants present EFNs). Paired t-tests were applied to assess whether the average difference in number of ants between paired plants (*Anemopaegma* and nearest neighbor) was higher than expected by chance.

For the second set of observations, another 14 individuals of *A. album* and 24 new individuals of *A. scabriusculum* were tagged in each sampling locality. For each individual, we recorded the average number of EFNs per leaflet (4–6 leaflets sampled without detaching the leaves), the number of leaflets

with active EFNs per plant, and the number of ants per plant. We used multiple linear regressions to test whether the variables representing EFNs were associated with the number of ants encountered on each plant. This analysis also included the covariate 'time of the day when ants were collected' in order to control for this source of variation in the data.

Effect of ant-visitation on herbivory and plant reproduction

Ant-exclusion experiments were carried out to evaluate the defensive role of ant visits against herbivores and its influence on flower and fruit production in *A. album* and *A. scabriusculum*. Owing to differences on the spatial arrangement of plants of the two species of *Anemopaegma*, we used a paired experiment to *A. scabriusculum* (more clumped distribution) and non-paired experiment to *A. album* (more scattered distribution). Specifically, we used 48 paired plants (24 sample units) of *A. scabriusculum* and 25 scattered plants of *A. album.* For each plant pair, one randomly selected plant was used as control, while the other was used for the ant-exclusion experiment. Ants were prevented from reaching treated plants with Tanglefoot® (Tanglefoot Co., MI, USA), a sticky resin that was applied as a barrier at the base of each individual. All ants found on treated plants were removed at the time of resin application. In addition, plants that could serve as bridges of ants to treated plants were pruned. None of these procedures were applied to the control individuals so that ants could have free access to those individuals.

Total leaf area and area of damage were measured on previously marked leaves in the field using visual assessment with the aid of an acrylic sheet that included a 0.3 cm^2 grid. The leaves that were opposite to the undamaged leaves selected for the experiments were marked with thin colored wires. For this experiment, we did not use the marked leaves for herbivory measurements in order to prevent the marking from having any effect on insect movement. Herbivory was estimated as a proportion of the leaf area that was removed during the first 15 days (first period), and during the following 55 days (second period; lasted between the 16th and the 70th experimental days). The leaf herbivory generated in the first sampling period was subtracted from the accumulated herbivory during all experimental days in order to assess herbivory in two different periods. To quantify herbivory in the first 15 days (first period), two kinds of marked leaves based on the leaf edge were used: 4–6 young leaflets and 4–6 mature leaflets per plant. Young leaves were marked when they were still unexpanded buds (or in the process of expansion), while mature leaves were marked after they were already fully expanded (and completely undamaged). Mature leaves were not followed during the second period (after the first 15 days of experiment), because most leaves showed signs of senescence.

Two separate analyses were performed with the data collected. The first analysis was a two-way ANOVA that was employed to test for the effect of leaf age (young vs. mature leaves) and ant presence (absence vs. presence) on the amount of herbivory accumulated in the first period. In this analysis, leaf age was included as an intrinsic factor of plants in order to control for this source of variation given that this factor has been previously associated with herbivory (i.e., higher herbivory levels were found in younger leaves; Coley 1980). The second analysis was employed to test for the effect of ants on herbivory through time. This analysis was a repeated-measures ANOVA with one variable between subject effects (ant exclusion) and one variable within subject effect (herbivory time period). Similarly, a repeated-measures ANOVA was also used to evaluate the impact of ants on the number of flowers produced per plant during the two sampling periods. Furthermore, the impact of ants on fruit set was evaluated through the difference in the number of fruits produced between control plants and plants in which ants had been removed. All analyses included the covariate "number of leaves of each plant" in order to control for the potential effect of plant size on herbivory.

Results

Extrafloral nectaries in *A. album* and *A. scabriusculum* are usually in depressions that are located in both surfaces of leaflets (Fig. 1, but see Appendix 2S in electronic supplementary material for a morphological description of EFNs). In *A. scabriusculum* these nectaries are more predominant in the adaxial surface, where they are scattered and present an average of

11.3 EFNs (SE = 0.8, $n = 24$ plants) per leaflet (Fig. 1b), with the abaxial surface with less than 1 EFN per leaflet. In *A. album,* plants presented an average of 3.6 scattered EFNs per leaflet (SE = 0.8, $n = 14$ plants) and 29.6 aggregated EFNs per leaflet. These EFNs were concentrated in the abaxial surface (SE = 4.3, $n = 14$ plants), forming clusters near the central vein at the base of the blade (Fig. 1d).

Histochemical analyses indicated that the secretory tissues of the EFNs from both species of *Anemopaegma* presented sugars, proteins, and lipids (Fig. 1e). However, no pectin/mucilage, phenolic compounds, or alkaloids were detected. Direct measurements on the exudates conducted with a portable refractometer (and test strips) detected a concentration of 71.0% (SE = 7.4, $n = 10$ plants with 37 leaflets in each one) sugar on 0.8 µl of the total exudate collected in *A. album* and 30.6% (SE = 3.6, $n = 21$ plants with 52 leaflets in each one) sugar on 27.3 µl of the total exudate in *A. scabriusculum*. The volume of nectar secreted by the EFNs was lower (and more concentrated) in *A. album* than in *A. scabriusculum*.

Four genera of ants dominated the interaction with EFNs (Table 1), with ants belonging to the genus *Camponotus* being the most abundant of the two species of *Anemopaegma*. On average, *A. album* had 12.4 ants/plants (SE = ± 3.3; $N = 14$) from which 20.5% were *Camponotus*, while *A. scabriusculum* had 23.3 ants/plants (SE = ±5.5; $N = 14$) from which 97% were *Camponotus*. The most common herbivores (Table 2) were the following: *Charidotis* sp. (Chrysomelidae, Coleoptera), *Aerenicopsis* sp. (Cerambycidae, Coleoptera), *Sumitrosis* sp. (Chrysomelidae, Coleoptera), and *Xestotrachelus robustus* (Romaleidae, Orthoptera). Observations of the interactions (physical contact) between ants and herbivores showed that all four common herbivores were indifferent to the most common ants of the genera *Camponotus* and *Cephalotes*.

No significant difference was encountered between the abundance of ants on individuals of *A. album* and neighboring species (Table 3). However, individuals of *A. scabriusculum* presented 11.7 more ants than their closest neighbors on average. Both *A. album* and *A. scabriusculum* presented a higher abundance of ants on plants with higher number of leaves with active EFNs (Fig. 2a, c). In *A. album*, every 30 leaflets (ten leaves) with active EFNs led to an increase of two ants per experimental plant. No ants were found in individuals without active EFNs (e.g., with senescent leaves or leaves at very young stages). Similarly, every 30 leaflets (ten leaves) with active EFNs of *A. scabriusculum* led to an increase of 1.8 ants per experimental plant. Furthermore, every five EFNs per leaflet of *A. album* led to an increase of 1.4 ants per plant on average (Fig. 2b). In *A. scabriusculum*, on the other hand, ant visitation did not vary with the average number of EFNs (Fig. 2d).

Significant differences were found between the damage of leaflets of young and mature leaflets of *A. album* ($F_{(1,22)} = 18.9$, $P < 0.001$) and *A. scabriusculum* ($F_{(1,21)} = 5.4$, $P = 0.030$) during the first 15 experimental days. During this period, control individuals of *A. album* presented an average leaflet damage of 21.1% (SE = 7.1, $n = 12$) on young leaflets, and an average leaflet damage of 1.1% (SE = 0.4, $n = 12$) on mature leaflets. Leaflets of *A. scabriusculum* were significantly less damaged during the same period; on average, 0.003% (SE = 6.8×10^{-4}; $n = 23$) of the total area of young leaflets, and 0.0002% (SE = 6.8×10^{-5}, $n = 23$) of the total area of mature leaflets were consumed. Considering the total foliar damage accumulated over the 70 experimental days (using only data from young marked leaves) the plants of *A. album* presented an average leaflet damage of 58.6% (SE = 11.3, $n = 12$), while *A. scabriusculum* presented an average leaflet damage of 39.4% (SE = 7.0, $n = 23$) of its area during the same period.

No difference was found for leaf damage between individuals with or without ants of *A. album* ($F_{(1,22)} = 0.12$, $P = 0.73$) and *A. scabriusculum* ($F_{(1,21)} = 1.61$, $P = 0.22$) in the two sampling periods (Fig. 3). Behavioral observations of the ants that were visiting EFNs (mostly *Camponotus*) in both species of *Anemopaegma* studied did not indicate any antagonistic interactions with the species of herbivores that may have contributed to a decrease in leaf damage (Table 3). The feeding behavior of the common herbivores, three species of beetles on *A. album* and a species of cricket on *A. scabriusculum*, showed a potential preference for feeding on individuals of *Anemopaegma* versus other plants in the field (A. Nogueira, personal observation). This result was stronger in three out of four herbivore species, in which both the larvae and adults fed on individuals of *Anemopaegma* (Fig. 1S in electronic supplementary material, Appendix 3S Table). Throughout

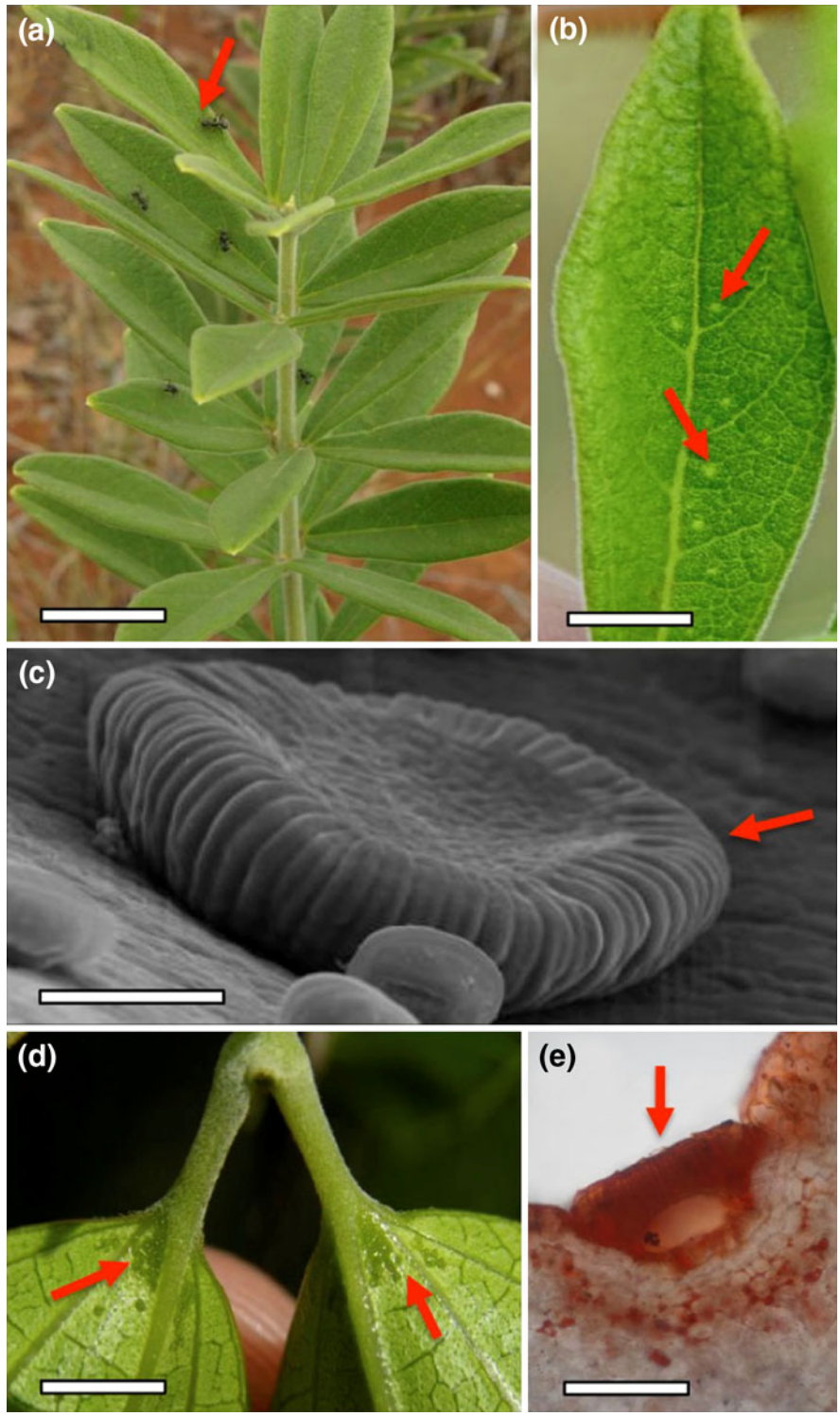


Fig. 1 Extrafloral nectaries of the *Anemopaegma* species. **a** Ants of the genus *Camponotus* visiting EFNs of *A. scabriusculum* (*scale bar* 40 mm); **b** EFNs scattered on the adaxial side of the leaflets of *A. scabriusculum* (*scale bar* 15 mm); **c** Detailed morphology of EFN of *Anemopaegma album* by SEM (*scale bar* 50 μm); **d** EFNs grouped at the axils of veins on the base of the abaxial side of leaflets of *A. album* (*scale bar* 10 mm); and **e** light microscopy of EFNs showing the positive reaction of total lipids of *A. album* (*scale bar* 140 μm)

the development of this study, only crickets (rarely) were seen feeding on plant species other than *Anemopaegma*.

Apart from the lack of an effect on herbivory in both species of *Anemopaegma* studied, ant visitation had no impact on flower ($F_{(1,22)} = 0.86$, $P = 0.36$)

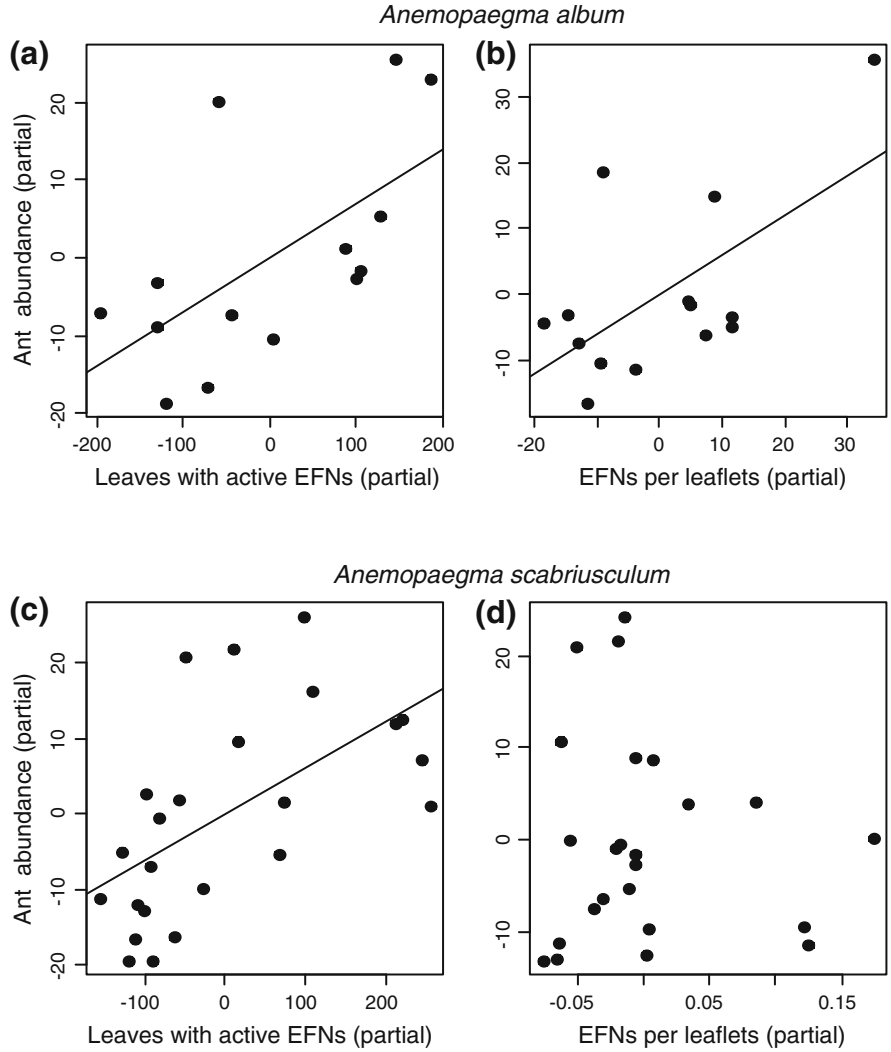


Fig. 2 Linear partial regressions of ant abundance and two different descriptors of EFNs in *A. album* ($F_{(2,11)} = 4.58$; $R^2 = 0.35$; $P < 0.04$) and *A. scabriuculum* ($F_{(2,21)} = 5.29$; $R^2 = 0.27$; $P < 0.02$). In *A. album,* ant abundance was positively correlated with the number of leaflets with active EFNs ($P = 0.02$), and the average number of EFNs per leaflet ($P = 0.02$). In *A. scabriusculum*, ant abundance was correlated only with the leaflets with active EFNs ($P = 0.01$), and not with the average number of EFNs per leaflet ($P = 0.68$)

Table 1 Percentage of plants occupied by the ants (per genus) that visited the EFNs of species of *Anemopaegma* and their respective behavior

Ant genus using nectar from EFNs	*A. album* ($N = 14$)	*A. scabriusculum* ($N = 18$)	Observed ant behavior
Camponotus	64.3%	100%	Individuals that move quickly, with intermediate abundance and recruitment, chemical defense.
Cephalotes	35.7%	5.5%	Individuals that move slowly, with low abundance and recruitment, without chemical defense
Crematogaster	7.1%	0%	Individuals that move slowly, with high abundance and recruitment, chemical defense
Pseudomyrmex	50%[a]	0%	Individuals that move quickly, with low abundance and recruitment, without chemical defense

[a] In general present with only one individual on the whole plant

Table 2 Herbivores most commonly seen feeding on vegetative portions (mainly leaves) of *A. album* and *A. scabriusculum*

Herbivore species	Species of *Anemopaegma*	Observed behavior of herbivores and damage to the plant
Beetle *Charidotis* sp. (Fig. 1S: A, B, C)	*A. album* (rarely in *A. scabriusculum*)	All life stages feed on the plant; adults were never seen feeding on or laying eggs on other plant species
Beetle *Aerenicopsis* sp. (Fig. 1S: D, E)	*A. album* (exclusively)	Only adult were observed feeding on plants; adults were exclusively seen feeding
		On the younger branches and midrib of the expanding leaflets
Beetle *Sumitrosis* sp. (Fig. 1S: F)	*A. album* (exclusively)	All life stages feed on the plant; larval stages were herbivore miners, and adults were never seen feeding on other plant species
Cricket *Xestotrachelus robustus* (Fig. 1S: G, H)	*A. scabriusculum* (rarely in *A. album*)	Young nymphs and adults were observed feeding on leaves; the younger the crickets, the higher the number of individuals observed on the plant. It was the only species seen also feeding on other plant species

Figure 1S available online in the supporting information

Table 3 Difference in ant abundance between individuals of *A. album* (or *A. scabriusculum*) and their nearest-neighbors

Paired experimental plants	Mean difference in ant abundance (±SD)	t-test
A. album × neighbors	3.8 ± 21.0	t paired$_{(1,13)} = 0.65$; $P = 0.25$
A. scabriusculum × neighbors	11.7 ± 18.7	t paired$_{(1,17)} = 2.66$; $P = 0.01$

Experiments were carried out on *A. album* ($n = 14$) and *A. scabriusculum* ($n = 18$) separately

and fruit production ($F_{(1,22)} = 0.03$, $P = 0.86$) in *A. album*, suggesting a lack of a defensive role by the ants that visited this species (Fig. 4a). Conversely, ants led to a decrease in flower production ($F_{(1,21)} = 11.03$, $P = 0.003$) in *A. scabriusculum*, suggesting a potential cost associated with the ant-EFN interaction in this species (Fig. 4b). Unfortunately, we were unable to evaluate the impact of ant visitation to fruit production in *A. scabriusculum* because only one individual (out of the 48 individuals sampled) produced fruits.

Discussion

This study evaluated the potential impact of EFNs and associated ants to prevent leaf damage, and consequently, to improve flower production and fruit-set in two species of *Anemopaegma* (Bignonieae, Bignoniaceae). We found that EFNs produce a secretion composed of sugars, and potentially proteins and lipids, all of which represent important components of the ant diet (Carroll and Janzen 1973). This nectar resource is especially important to ant groups with some specialized anatomy (e.g., proventriculus adapted to benefit from the availability of liquid foods, Davidson 1997), such as that encountered in *Camponotus*, the most abundant group of ants sampled on the EFNs of the two species of *Anemopaegma* studied. *Camponotus* ants were also seen dominating the interaction with plants in the savannas (Oliveira et al. 1987; Del-Claro et al. 1996; Oliveira 1997; Korndorfer and Del-Claro 2006; Nascimento and Del-Claro 2010; Byk and Del-Claro 2010). In those studies, *Camponotus* ants were also the most abundant, as well as showed potentially protective patterns against herbivores (mostly Lepidoptera). On the other hand, ants belonging to the genus *Cephalotes* (the second most abundant in our study) did not present any defensive role for plants and were likely commensal or parasitic (see Byk and Del-Claro 2010). Studies addressing the role of each ant group in facultative ant–plant interactions mediated by EFNs only started to be conducted experimentally very recently (e.g., Byk and Del-Claro 2010). In the particular case of Neotropical savannas, it is reasonable to assume that *Camponotus* might play an important role in these interactions.

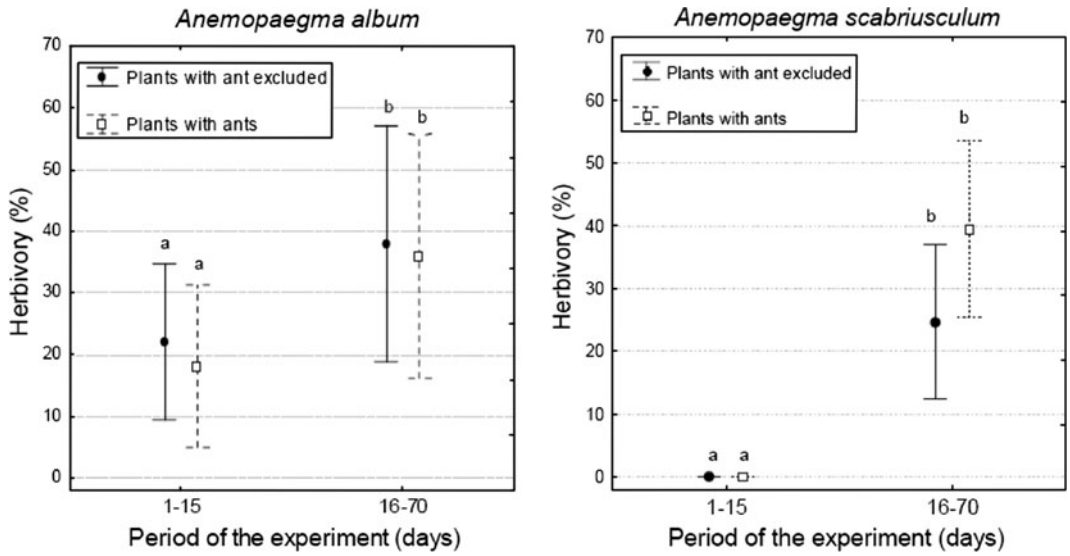


Fig. 3 Leaflet damage during the two sampling periods (1–15 days and 16–70 days). No effect of ant visitation was observed on leaf damage of *A. album* or *A. scabriuculum*. No significant interaction between factors was found, indicating that the effect of ant exclusion on the level of herbivory was consistent between sampling periods. Similar *letters* on vertical bars (0.95 confidence intervals) indicate no significant difference within each sampling period

In this study, EFNs led to an increase in ant visitation in both species of *Anemopaegma*. However, contrary to expectations under traditional plant-defense hypotheses, none of these plant species were defended by ants (Beattie 1985; Heil and McKey 2003; Rico-Gray and Oliveira 2007). One possible scenario is that the EFNs were not able to attract the minimum amount of ants that is necessary to reduce leaf herbivory. Studies conducted in savannas documented an average of 4.42 (±4.3) ants/plant in *Qualea grandiflora* (Oliveira et al. 1987), and an average of 1.97 (±1.6; daytime) and 1.80 (±1.6; nighttime) ants/plant in *Caryocar brasiliense* (Oliveira 1997), both of which were shown to represent sufficient amounts of ants to protect those taxa against herbivores. In our study, an average of 12 ants/plant was documented in *A. album*, and an average of 23 ants/plant in *A. scabriusculum*. Even if the studied individuals were controlled for plant size, the studied species of *Anemopaegma* would still present a higher number of ants per plant than *Q. grandiflora* and *C. brasiliense* suggesting that factors other than ant number (e.g., the identity of the ants or herbivores), would be responsible for the 'inefficiency' of the nectaries.

The cost-benefit model can be applied for the study of traits that are associated with the ant–plant-herbivore system (Heil and McKey 2003). Under this model, the production of EFNs and associated secretions are costly to the plant, however, this cost is outweighed by the benefits provided by ant visitors, leading to an increase in plant-fitness (Heil and McKey 2003). However, other scenarios are also possible for these interactions (e.g., commensalism or parasitism with ants; context-dependency by Jones and Callaway 2007). More specifically, ants may not represent effective defenses to the plants, and/or herbivores may be more efficient than expected at feeding the plants (e.g., Tempel 1983; Kersch and Fonseca 2005). Following this reasoning, it is possible that the community of ants and herbivores found in this study might explain the ineffective anti-herbivore defense of the nectaries. In fact, further information in ten populations of *A. album*, suggests that both ant and herbivore assemblage substantially vary across population, as well vary the EFNs abundance and the leaf damage by herbivores (A. Nogueira et al., unpublished). Moreover, we did not find a defensive role of ants, nor we found a positive impact to plant reproduction in both species of *Anemopaegma*. In *A. scabriusculum* in particular, we actually found a negative impact of ants for plant reproduction. These results suggest that the costs and benefits of EFNs might be equivalent in *A. album* (ants as commensal) but more costly than beneficial in *A. scabriusculum* (ants as parasites).

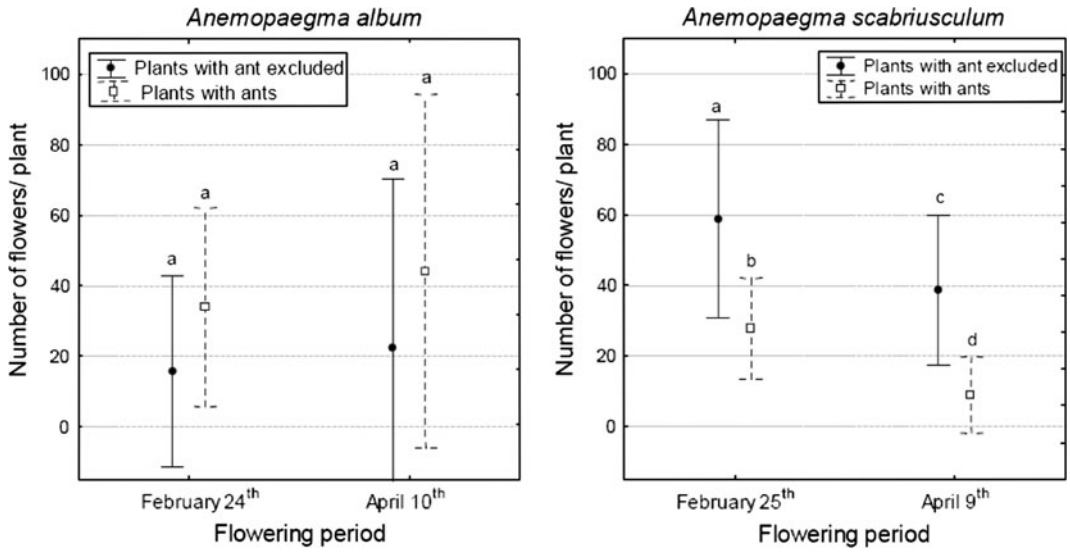


Fig. 4 Flower production in *A. album* (*left*) and *A. scabriusculum* (*right*). Flower production did not vary with ant visitation in *A. album*. In *A. scabriusculum,* flower production decreased with ant visitation. Results for each species were similar in the two sampling periods (February and April, 2009; that is the interaction between factors was not significant). Different *letters* on vertical bars (0.95 confidence intervals) indicate a significant difference within each sampling period

Apart from the cost-benefit model, the hypothesis of phylogenetic inertia might also explain the lack of trait functionality in some species or populations. In particular, phylogenetic inertia might play an important role in lineages in which the history of habitat occupation by taxa varied between different selective regimes (e.g., groups of closely related species in which some species occur in forests and others in savannas). Under this scenario, traits found in the ancestors of particular lineages would be maintained in their descendants due to phylogenetic inertia (see Orzack and Sober 2001; Blomberg and Garland 2002). In the specific case of EFNs, phylogenetic inertia would imply the maintenance of EFNs in some taxa simply because EFNs were present in the ancestor of the lineage and not necessarily because EFNs would confer higher fitness to these taxa presently or in new habitats. This hypothesis was initially used to explain the occurrence of plants with EFNs in an island of the Hawaiian archipelago that lacks ants (Keeler 1985).

In the particular case of *Anemopaegma*, the evolutionary history and morphological characteristics of the studied species might help explain the presence of EFNs in spite of its inefficient defensive role. Indeed, EFNs were already present in the ancestors of *Anemopaegma* (A. Nogueira et al., unpublished, Fig. 5). Furthermore, EFNs were likely more abundant in the ancestors than in the studied species (A. Nogueira et al., unpublished), those of which occupied forest habitats (Lohmann 2003; Lohmann 2011, in press). EFNs were subsequently kept in all species of *Anemopaegma* even after lineages occupied drier areas such as the Brazilian savannas. It is possible that these structures might have presented a defensive role in the past but that this role was lost after the transition to dry areas, where these species might have started to interact with a new assemblage of ants and herbivores. Under the new selective regime, species with EFNs might only experience the cost associated with the production of EFNs but not the benefits (defensive role), leading to an overall decrease in fitness surrogates such as flower and fruit production. Indeed, the average abundance of EFNs is higher in species of *Anemopaegma* that are currently distributed through forests than in species that are currently distributed through savannas (see Appendix 3S in electronic supplementary material), suggesting a decrease in the number of EFNs after the invasion of savannas. However, this study still lacks information on the role of the EFNs in forest species, which would be very important in order to test the phylogenetic inertia hypothesis.

Another non-mutually exclusive explanation for the lack of defensive role of EFNs in the studied

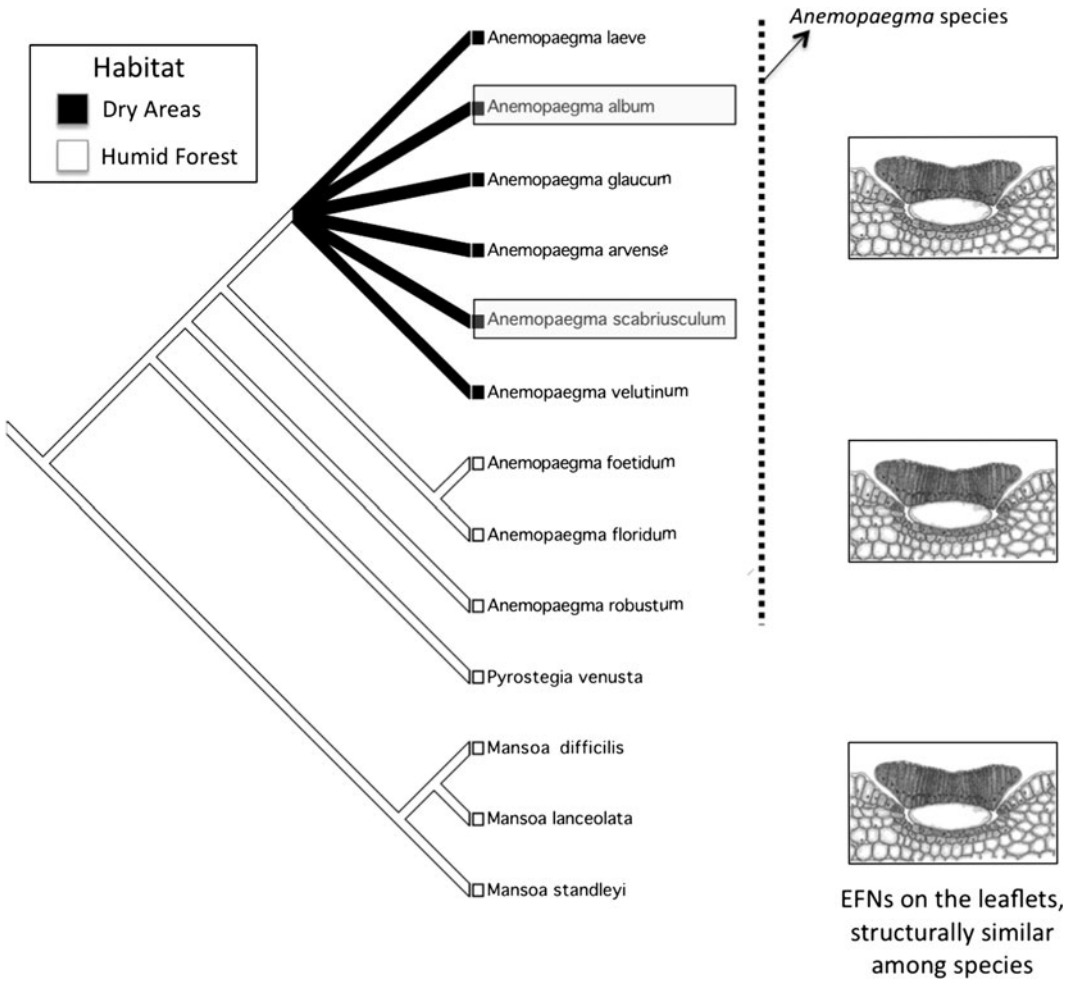


Fig. 5 Phylogenetic relationships between species of *Anemopaegma* (modified from Lohmann 2006; Lohmann et al., unpublished) indicating that the ancestor of *Anemopaegma* occupied a humid environment and already presented EFNs; subsequent lineages moved into dry areas and maintained the EFNs

species falls within the framework of the geographic mosaic theory of coevolution (GMT) (Thompson 2005). The GMT has incorporated a geographical component for the study of interactions by remarking that different populations from a single species might be under different biotic and abiotic pressures (context-dependency of the interaction outputs differently in each population). Within the context of our study, GMT would predict local differences among populations (and species in expanded temporal scale), which might lead to variation in the ant–plant interaction (e.g., Alonso 1998; Manzaneda and Rey 2008), with some populations not being protected by the ants. The geographic variation of ant–plant interactions on savannas were observed among populations of the ant-plant *Tococa guianensis*, in which some populations were found to persist in some areas without these obligate ants on the plants (Moraes and Vasconcelos 2009). The lack of a defensive role of EFNs in both species of *Anemopaegma* studied suggests that the ants available in our study sites were not sufficiently aggressive to protect the visited plants against herbivores (very efficient herbivores). Indeed, none of the ants observed visiting populations of *A. album* presented an aggressive behavior against the most common herbivores encountered in these species. Among the herbivores observed in *A. album*, adults and larvae of *Charidotis* sp. (Cassidini, Chrysomelidae, Coleoptera) were the ones that contributed for the highest amount of leaf damage. Field observations

indicated that common ants of the genus *Camponotus* and secondarily *Cephalotes* did not interfere in the reproduction or oviposition of adults, nor in the movement of larvae of the herbivore *Charidotis* sp. A similar pattern was observed in the other two most common herbivores that visited *A. album*. In *A. scabriusculum*, on the other hand, the highest amount of leaf damage was caused by a single herbivore, the cricket *Xestotrachelus robustus*. Similar to the patterns observed in *A. album*, the ants observed in *A. scabriusculum* (mostly *Camponotus*) did not seem to protect plants from cricket damage.

Overall, our results suggest that plants with higher densities of EFNs might not present an advantage over individuals with lower densities of EFNs in populations of *A. album* and *A. scabriusculum*. However, it remains unclear why exactly nectaries remained present in these populations. It is possible, for instance, that EFNs might have remained as a result of some specific predictions associated with the GMT (Thompson 1999) such as: (1) source populations efficiently defended by ants (i.e., "hotspots") might be supplying the genetic variation necessary for the presence and abundance of EFNs through gene flow in populations not efficiently defended by ants (i.e., "coldspots"), (2) momentarily inefficient populations might be defended from time to time by ants (temporal variation of the system), leading to a maintenance of EFNs even during the periods in which these structures were inefficient, (3) the cost of EFNs and its respective secretion would be very low in savanna-like environments (usually with abundant light and a large reserve of carbohydrates in plants), maintaining the variation of EFNs even though these structures would be inefficient or ineffective in reducing herbivory under such conditions (almost like a "neutral" trait).

Although the temporal and spatial scales of this study are restrictive to unfold the exact causes of the patterns encountered, it is reasonable to assume that the conditionality of the interactions is associated with the local community of ants and herbivores, and indirectly with the variation of abiotic factors. Furthermore, it is also reasonable to hypothesize that the pattern encountered might have resulted from a loss of the functionality of EFNs that likely happened following the transition of taxa from humid forests to savannas. In order to fully test this hypothesis, it would be important to conduct studies within a broad geographic context. More specifically, it would be important to gain a deeper understanding of the patterns of variation of EFNs (and the role of them) found in taxa that presently occur in forests and savannas, using other representatives of the plant family Bignoniaceae in which similar habitat transitions and morphological structures are present. Only with a broader sampling, including detailed studies with multiple lineages and populations will we be able to gain a more thorough understanding of the ecological and evolutionary processes that shaped the evolution of EFNs in those species.

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References

Alonso LE (1998) Spatial and temporal variation in the ant occupants of a facultative ant-plant. Biotropica 30: 201–213

Beattie AJ (1985) Plant protection by direct interaction. The evolutionary ecology of ant-plant mutualisms. Univ of Cambridge, Cambridge, pp 21–53

Blomberg SP, Garland T (2002) Tempo and mode in evolution: phylogenetic inertia, adaptation and comparative methods. J Evol Biol 15:899–910

Bronstein JL (1994) Conditional outcomes in mutualistic interactions. TREE 9:214–217

Byk J, Del-Claro K (2010) Nectar- and pollen-gathering Cephalotes ants provide no protection against herbivory: a new manipulative experiment to test ant protective capabilities. Acta Ethol 13:33–38

Carroll CR, Janzen DH (1973) Ecology of foraging by ants. Annu Rev Ecol Syst 4:231–257

Chamberlain SA, Holland JN (2009) Quantitative synthesis of context dependency in ant-plant protection mutualisms. Ecology 90:2384–2392

Christianini AV, Oliveira PS (2010) Birds and ants provide complementary seed dispersal in a Neotropical savanna. J Ecol 98:573–582

Coley PD (1980) Effects of leaf age and plant life history patterns on herbivory. Nature 284:545–546

Costa AN, Vieira-Neto Vasconcelos HL, EHM Bruna EM (2008) Do herbivores exert top-down effects in Neotropical

savannas? Estimates of biomass consumption by leaf-cutter ants. J Veget Sci 19:849–854
Crist TO, Macmahon JA (1992) Harvester ant foraging and shrub steppe seeds—interactions of seed resources and seed use. Ecology 73:1768–1779
Davidson DW (1997) The role of resource imbalances in the evolutionary ecology of tropical arboreal ants. Biol J Linn Soc 61:153–181
Del-Claro K, Berto V, Réu W (1996) Effect of herbivore deterrence by ants on the fruit set of an extrafloral nectary plant, *Qualea multiflora* (Vochysiaceae). J Trop Ecol 12: 887–892
Eiten G (1972) The cerrado vegetation of Brazil. Bot Rev 38:201–341
Elias TS, Gelband H (1975) Nectar: its production and functions in trumpet creeper. Science 189:289–291
Heil M, McKey D (2003) Protective ant-plant interactions as model systems in ecological and evolutionary research. Annu Rev Ecol Evol Syst 34:425–453
Hickman JC (1974) Pollination by ants—low-energy system. Science 184:1290–1292
Janzen DH (1966) Coevolution of mutualism between ants and acacias in Central America. Evolution 20:249–275
Jones CG, Callaway RM (2007) The third party. J Veget Sci 18:771–776
Keeler KH (1985) Extrafloral nectaries on plants in communities without ants: Hawaii. Oikos 44:407–414
Kersch MF, Fonseca CR (2005) Abiotic factors and the conditional outcome of an ant-plant mutualism. Ecology 86: 2117–2126
Korndorfer AP, Del-Claro K (2006) Ant defense versus induced defense in *Lafoensia pacari* (Lythraceae), a myrmecophilous tree of the Brazilian cerrado. Biotropica 38:786–788
Lohmann LG (2003) Phylogeny, classification, morphological diversification and biogeography of Bignonieae (Bignoniaceae). Ph.D. thesis, University of Missouri-St. Louis, St. Louis
Lohmann LG (2006) Untangling the phylogeny of Neotropical lianas (Bignonieae, Bignoniaceae). Am J Bot 93:304–318
Lohmann LG (2011) A new generic classification of Bignonieae (Bignoniaceae) based on molecular phylogenetic data and morphological synapomorphies. Ann Miss Bot Gard (in press)
Machado SR, Morelatto LPC, Sajo MG, Oliveira PS (2008) Morphological patterns of extrafloral nectaries in woody plant species of the Brazilian cerrado. Plant Biol 10: 660–673
Manzaneda AJ, Rey PJ (2008) Geographic variation in seed removal of a myrmecochorous herb: influence of variation in functional guild and species composition of the disperser assemblage through spatial and temporal scales. Ecography 31:583–591
Moraes SC, Vasconcelos HL (2009) Long-term persistence of a Neotropical ant-plant population in the absence of obligate plant-ants. Ecology 90:2375–2383
Nascimento EA, Del-Claro K (2010) Ant visitation to extrafloral nectaries decreases herbivory and increases fruit set in *Chamaecrista debilis* (Fabaceae) in a Neotropical savanna. Flora (in press)
Ness JH (2003) *Catalpa bignonioides* alters extrafloral nectar production after herbivory and attracts ant bodyguards. Oecologia 134:210–218
Oliveira PS (1997) The ecological function of extrafloral nectaries: herbivore deterrence by visiting ants and reproductive output in *Caryocar brasiliense* (Caryocaraceae). Funct Ecol 11:323–330
Oliveira PS, Brandão CRF (1991) The ant community associated with extrafloral nectaries in the Brazilian cerrados. In: Huxley CR, Cutler DF (eds) Ant-plant interactions. Oxford University Press, Oxford, pp 198–212
Oliveira PS, Freitas AVL (2004) Ant-plant-herbivore interactions in the Neotropical cerrado savanna. Naturwissenschaften 91:557–570
Oliveira PS, Leitão-Filho HF (1987) Extrafloral nectaries—their taxonomic distribution and abundance in the woody flora of cerrado vegetation in southeast Brazil. Biotropica 19:140–148
Oliveira PS, Silva AF, Martins AB (1987) Ant foraging on extrafloral nectaries of *Qualea grandiflora* (Vochysiaceae) in cerrado vegetation: ants as potential antiherbivore agents. Oecologia 74:228–230
Orzack SH, Sober E (2001) Adaptation, phylogenetic inertia, and the method of controlled comparisons. In: Orzack SH, Sober E (eds) Adaptationism and optimality. Cambridge University Press, Cambridge, pp 45–63
Pirani JR, Mello-Silva R, Giulietti AM (2003) Flora de Grão-Mogol, Minas Gerais, Brasil. Boletim de Botânica da Universidade de São Paulo 21:1–24
Ratter JA, Ribeiro JF, Bridgewater S (1997) The Brazilian cerrado vegetation and threats to its biodiversity. Ann Bot 80:223–230
Rico-Gray V, Oliveira PS (2007) The ecology and evolution of ant-plant interactions. University of Chicago Press, Chicago
Rosumek FB, Silveira FAO, Neves FS, Barbosa NPU, Diniz L, Oki Y, Pezzini F, Fernandes GW, Cornelissem T (2009) Ants on plants: a meta-analysis of the role of ants as plant biotic defenses. Oecologia 160:537–549
Stephenson AG (1982) The role of extrafloral nectaries of *Catalpa speciosa* in limiting herbivory and increasing fruit production. Ecology 63:663–669
Tempel AS (1983) Bracken fern (*Pteridium aquilinum*) and nectar-feeding ants: a nonmutualistic interaction. Ecology 64:1411–1422
Thompson JN (1999) Specific hypotheses on the geographic mosaic of coevolution. Am Nat (suppl.) 153:S1–S14
Thompson JN (2005) The geographic mosaic of coevolution. Univ of Chicago Press, Chicago